EVERY F-ALGEBRA ENDOMORPHISM OF AN ALGEBRAIC FIELD EXTENSION IS AN AUTOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The argument that f maps the root set of P into itself is the point where f being an F-algebra map is used (P has coefficients in F).

Editorial note. The label lemma-algebraic-extension-self-map names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-field-maps-injective (a field map over the base is injective), lemma-algebraic-finitely-generated (an algebraic finitely generated extension is finite)

Source reference. Original reference 09GH: sources/stacks-fields/pinned-original-source.tex, source lines 910–910. Prerequisite pointer invoked at 1038; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09FU: sources/stacks-fields/pinned-original-source.tex, source lines 471–471. Prerequisite pointer invoked at 1031; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/F an algebraic extension of fields. Any F-algebra map $f : E \to E$ is an automorphism.*

Proof. If E/F is finite, then $f : E \to E$ is an F-linear injective map (Lemma 09FU) of finite dimensional vector spaces, and hence bijective. In general we still see that f is injective. Let $\alpha \in E$ and let $P \in F[x]$ be a polynomial such that $P(\alpha) = 0$. Let $E' \subset E$ be the subfield of E generated by the roots $\alpha = \alpha_1, \ldots, \alpha_n$ of P in E. Then E' is finite over F by Lemma 09GH. Since f preserves the set of roots, we find that $f|_{E'} : E' \to E'$. Hence $f|_{E'}$ is an isomorphism by the first part of the proof and we conclude that α is in the image of f. □

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